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Carbon pricing versus emissions trading: A supply chain planning perspective

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ABSTRACT

Carbon pricing (taxes) and carbon emissions trading are two globally practiced carbon regulatory policy schemes. This paper presents an analytical supply chain planning model that can be used to examine the supply chain performance at the tactical/operational planning level under these two policy schemes. Model implementation and analyses are completed using actual data from a company operating in Australia, where these environmental regulatory policies are practiced. Numerical results provide important managerial and practical implications and policy insights. In particular, the results show that there are inflection points where both carbon pricing and trading schemes could influence costs or emissions reductions. An erratic nonlinear emissions reduction trend is observed in a carbon pricing scheme as the carbon price increases steadily; whereas emissions reduction in a carbon trading scheme follows a relatively linear trend with a nonlinear cost increase. Overall, a carbon trading mechanism, although imperfect, appears to result in better supply chain performance in terms of emissions generation, cost, and service level; even though a carbon tax may be more worthwhile from an uncertainty perspective as emissions trading costs depend on numerous uncertain market conditions.

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1. Introduction

Environmentally sustainable supply chain (SC) planning, also termed green SC planning, aims to develop unified design, planning and optimization models in which economic goals such as profit maximization and cost minimization are integrated with environmental goals such as carbon and greenhouse gas emissions minimization (Sundarakani et al., 2010; Varsei et al., 2014). The adoption of green SC planning efforts is greatly influenced by two widely-practiced regulatory efforts including carbon pricing (taxing) and carbon trading schemes (Schaltegger and Csutora, 2012).

SC planning and optimization on its own is a relatively intricate process with numerous variables and constraints to be taken into consideration and the incorporation of environmental dimensions adds to its complexity (Fahimnia et al., 2014a). Organizations facing these complex decision environments can find utility in tools for planning and managing their SCs. The development of SC modeling tools that have effectively integrated and evaluated environmental issues, alongside economic and business concerns, have only started to receive significant interest (Benjaafar et al., 2013; Brandenburg et al., 2014; Seuring, 2013; Tang and Zhou, 2012). Many of these modeling efforts focus on strategic planning levels of analysis such as the design of SC networks, while challenges at the tactical and operational planning levels are less explored (Seuring, 2013). In fact, the developments in some areas such as reverse logistics have dominated the early and recent green SC modeling literature (Srivastava, 2007).

Motivated by actual regulatory climate change pressures that are evolving in Australia, we develop and apply an analytical planning model to explore how organizations can manage their SCs under two carbon regulatory schemes. Not only are practical implications associated with the modeling effort presented, but research implications including further model development and investigations of additional outcomes are thoroughly discussed in this paper. The primary objective of this work focuses on the development and analysis of SC planning under emergent regulatory regimes. The proposed SC planning model contributes to the green SC modeling literature through helping organizations, policymakers, and even NGO's evaluate the tactical and operational implications from broad-based regulatory policy decisions.

The remainder of the paper is organized as follows. We begin in Section 2 with some background on environmental regulatory policies and organizational responses to these policies. Green SC management modeling efforts specifically those with a clear focus

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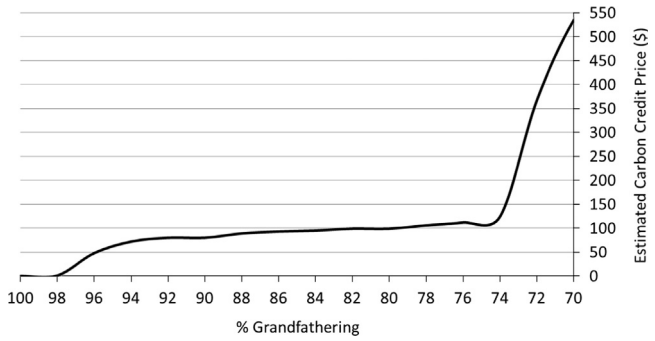

Fig. 3. Estimated carbon price to achieve grandfathering goals in the carbon trading scenario.

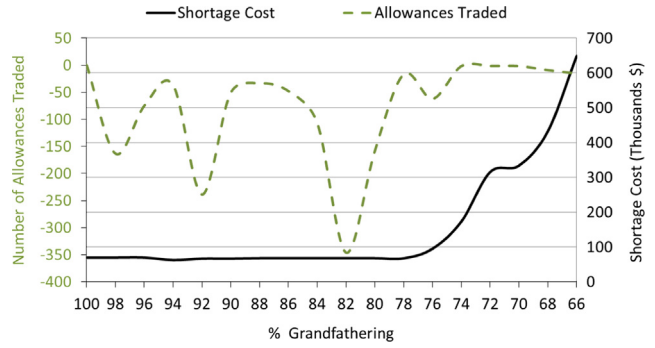

Fig. 4. Traded allowances against shortage cost in the carbon trading scenario.

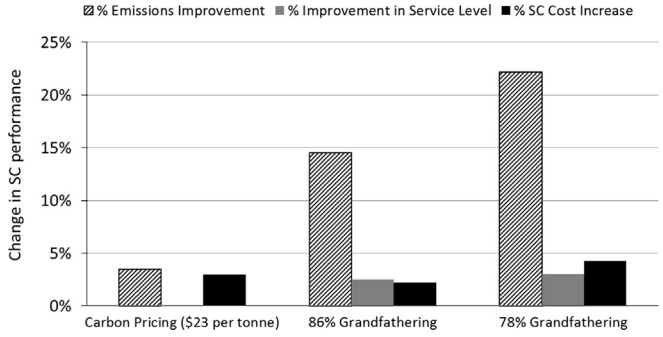

Fig. 5. SC performance under carbon pricing and carbon trading schemes.

benefits associated with the grandfathering level, especially when facing the nonlinear and erratic behavior of optimal cost solutions based on emissions reductions.

Using the fix carbon price trial-and-error procedure defined in Section 4.2, Fig. 3 shows the carbon prices required to achieve certain grandfathering goals. For instance, a carbon allowance price of $80 per tonne is required to force ABC to optimally reduce its carbon emission by 10% (i.e. a 90% grandfathering goal). It should be noted that the actual price of carbon allowances will be determined by the supply and demand for allowances in the market. The values in Fig. 3 are the estimated prices a company should be charged to keep its carbon emissions below the target grandfathering amounts. The actual market carbon price may be highly dependent on the performance of the other companies trading their allowances (i.e. how many surplus allowances are being sold and how many additional allowances are being purchased by affected companies). Governments and policymakers may also have influences on the actual carbon price through decisions on the provision of subsidies as well as the effective carbon cap setup. Fig. 3 shows that a 4% to 26% cap reduction from the baseline results in a slow, steady increase for the best price. But a cap of over a 26% reduction (grandfathering lower than 74%) will cause a very large carbon price increase and significant SC cost increase. It is at this stage that ABC should definitely consider investment in carbon emissions mitigation of its SC activities, either through SC restructuring or technology investment decisions. This result may not be true for all companies but a similar analysis can be completed for different cases.

One important policy question in a carbon trading environment can be how to determine the best percentage for the grandfathering level at which the actual market prices are close to the companies' estimated carbon credit price. It can be argued that carbon price estimations are more accurate if the number of allowances sold and the estimated revenue generated by a company is closer to zero. This level would have the minimal impact of the trading prices on SC performance. Fig. 4 shows a clear illustration of the number of traded allowances at each grandfathering level (the dashed line in Fig. 4). It can be seen that in some instances the number of allowances sold is close to the zero trading line, see for instance the 78% and 74–70% grandfathering levels. In another observation, when comparing the trends in traded allowances in Fig. 4 with numerical results in Table 2, we observe that the sharp decreases in the number of allowances traded at percentage grandfathering of 82% and 92% coincide with major reductions in transport emissions at these points through the use of greener transportation modes. These however occur in conjunction with increased transportation costs at these points.

This analysis should be examined against SC service level, an important SC performance measure. SC service level can be defined as penalties and compensation paid to customers in case of demand backordering. Fig. 4 indicates that the overall shortage cost at ABC increases dramatically after the 78% grandfathering point. Given the existing SC infrastructure, equipment, and technology, if emissions reduction is to be greater than 22%, ABC needs to either sacrifice its service level or increase its SC costs to a large extent. This type of decision can be examined against investment in alternative emissions mitigation strategies. This situation causes a conflict in the policymaker and industry practitioner perspectives. For example, Fig. 4 shows that the 74%, 72% and 70% grandfathering levels can be an ideal policy situation in terms of the number of allowances traded, with only 2 allowances (rounded up) traded in all cases, with 26%, 28% and 30% emissions reductions, respectively. However, from an organizational perspective, 74%, 72% and 70% grandfathering levels would imply an increase in overall SC cost by 6.72%, 12.07% and 14.65% (Fig. 2) as well as 148%, 353% and 379% poorer service levels respectively (i.e. $173,007, $315,106 and $333,804 shortage costs compared to a baseline of $69,627).

Fig. 5 provides a graphical comparison between the current carbon pricing situation in Australia (a $23 carbon price in Scenario 1) and two optimal points of Scenario 2 (in terms of SC cost, service level and number of allowances traded). Values are shown relative to the baseline situation when there is no carbon pricing or trading scheme in place. For Scenario 1, Fig. 5 shows the numerical results (emissions generation, service level and SC cost) for the current carbon price of $23 per tonne. From Scenario 2, we choose two situations, 86% and 78% grandfathering which have a reasonable SC cost and service level and require a small number of allowance trading, 48 and 17 allowances respectively (see Fig. 4).

ABC performs better in all aspects, emissions, service level, SC cost at the 86% grandfathering point when compared to the current carbon pricing scheme. The SC cost in this case is increased by about 2.2%, while carbon emissions and service level improve by 14% and 2.5% respectively. The 78% grandfathering point is where the environmental performance is improved to a marginally greater extent (i.e. 22% improvement), but the overall SC cost is accordingly increased noticeably (by about 4.25%). This result, if spread across all companies in an industry, may impose

considerable impact on the national economy and competitive positioning of firms. Overall, carbon trading may outweigh a carbon pricing scheme, but the carbon cap amount (the percentage grandfathering) is subject to many factors including the carbon reduction goals and the current economic situation. It may also depend on what alternatives exist to help fund new processes technologies, infrastructure, and equipment.

Although there is more flexibility with industry support for a market-based mechanism for carbon trading, a carbon fixed price (carbon tax) may actually be more beneficial from an uncertainty perspective. If companies know that the carbon price will be stable, they can make investments to improve their carbon emissions performance based on certain price data. But, one challenge of a fixed price is determining what price to set. A wrong carbon fixed price may either cause unnecessary burden if it is fixed too high or go in the other extreme and be so low that no reductions are actually made in carbon emissions, which may be the case for ABC. Of course, any revenue generated by the fixed price (tax) can be used to help invest in public efforts to reduce carbon emissions. Our findings show that a market-based carbon trading mechanism may be a more effective way, even though imperfectly, to set the price to achieve the reduction in carbon emissions.

5. Conclusions and future research

Many nations, in response to global climate change caused by greenhouse gas emissions, have introduced carbon reduction regulatory policy schemes. Carbon pricing and emissions trading are the two most widely practiced schemes. Companies will consider their SC operations as carbon emitters and use planning and optimization tools to help evaluate their cost and emissions performance under these carbon regulatory policies. Given this context, in this paper we linked environmental regulatory policies to internal SC planning practices using a tactical SC optimization model. Two scenarios reflecting the two environmental regulatory policies were run using actual company data. Results were analyzed from overall SC cost and emissions reduction perspectives.

The results showed that various planning and policy insights can be gathered with the proposed mathematical programming optimization model. In both circumstances, the carbon pricing and trading schemes, there seemed to be inflection points where certain policies or practices could more meaningfully influence costs or emissions reductions. We have seen erratic nonlinear emissions reduction trends in a carbon pricing scheme as carbon price increases steadily, whereas in a carbon trading scheme, emissions reduction follows a relatively linear trend, but the SC cost increase is rather nonlinear. While a carbon tax may be more beneficial from an uncertainty perspective, we find that a carbon trading mechanism, although imperfect, result in better supply chain performance in terms of emissions generation, cost and service levels. Although the results are for one company and their planning purposes, policymakers can broaden the model to include various industries and company inputs. Whether all industries and companies will have similar characteristics is not known at the time, but the utility of such a model for a single organization is quite clear.

There are certain limitations for this study, but these limitations also provide opportunities for further research. First, extending the model to industry level analyses can be completed, although it may require the incorporation of additional variables, constraints and assumptions. Using broad-based macro-economic input-output analysis may be one way of helping define data that can be used to judge the influences of different prices and policies. Another limitation of the technique was the need to complete a trial-and-error for some price setting determinations. Making carbon price parameter a decision variable, although it might cause some nonlinearities in the resulting models, may allow it to jointly determine the desired price that can force a company to reduce the emissions level below the carbon cap goal. Also, our modeling effort was at the tactical operational planning levels, and hence the infrastructure, technology, equipment choices, and processes were considered fixed. The incorporation of such strategic decisions into the developed model would allow for consideration of when to invest in improvements rather than being completely reactive to just purchasing or selling allowances.

The field of green SC planning and emergent climate change regulatory policy is still evolving. The complexities of the decisions and policy relationships require thoughtful modeling. We have shown the utility of the tactical SC planning model developed in this paper and how advances in thought and practice can be achieved in numerous directions. The common saying is that we can think globally (strategic design and planning to tackle climate change in the long-term), but act locally (tactical and operational planning, harmonized with those strategic goals). This research shows the necessary investigations needed at multiple levels of analysis to help us solve one of the humanity's most pressing concerns.

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